

Delivered tables — preview

Not a deliverable. Layout here is arbitrary; the paper's floats are D's call. Generated by `python -m analysis.preview`.

table1_dataset

Statistic	Development	Test
Paragraphs	2000	2000
Source reports (PDFs)	49	49
shared across splits	49	49
Companies	49	49
Legal states observed	15 / 17	n/a
<i>Rarest classes</i>		
within_2_years	34	n/a
Misleading	2	n/a

Dataset and split statistics, regenerated from the versioned data by `analysis/audit.py`. Misleading occurs twice in the whole development set, in two different reports, and is absent from the Calibration partition of 18 of the 30 rotations. The four fields admit 120 combinations, of which the hierarchy leaves 17 legal, and no development row violates it. All 49 development reports also appear in the test split, and one paragraph text is duplicated across the two. No company contributes more than one report, so a document-disjoint split is also company-disjoint. The competition test split ships no labels, so its label-derived cells are marked n/a.

table2_main

ID	Calibration	Decoding	Weighted F1	PS	VT	ES	EQ	Tuple Acc.	Invalid %
M0	None	Independent	0.572±0.003	0.796	0.464	0.650	0.424	0.359	12.6
M1	None	Projection	0.571±0.003	0.796	0.467	0.651	0.418	0.394	0.0
M2	Global	Projection	0.574±0.004	0.796	0.493	0.650	0.418	0.428	0.0
M3	Conditional	Projection	0.575±0.003	0.796	0.501	0.651	0.416	0.432	0.0
M4	None	17-state	0.571±0.005	0.799	0.468	0.648	0.420	0.388	0.0
M5	Global	17-state	0.576±0.001	0.796	0.493	0.649	0.423	0.430	0.0
M6	Conditional	17-state	0.567±0.008	0.784	0.494	0.634	0.418	0.427	0.0

Controlled comparison of decision rules on identical base probabilities and identical test rows. Each cell is the mean over three seeds of a single score computed on all 2,000 concatenated test rows; $\pm$ is the sample standard deviation across seeds, which reflects the variability of the whole pipeline – fold assignment and training together – rather than model stability. Paired contrasts between these rows, under each metric and with their Holm-corrected p-values, are reported in Table 4. Under that variant M0 scores 0.567 against 0.572 on the official metric, while M1–M6 score identically under both: only the unconstrained arm predicts fields its own ancestors do not support. The metrics disagree about the winner: M5 on the official metric and M6 on hF, which is the lowest-scoring rule on the official metric. A ranking is only meaningful stated together with the metric that produced it.

table3_legality_cost

Backbone	λ	M0 invalid	Enforce legality (M1–M0)		Joint decoding (M4–M1)	
			Official	Whole-row	Official	Whole-row
RoBERTa-large	0	12.55%	-0.001	+0.035	+0.000	-0.006
DeBERTa-v2-320M	0	19.75%	-0.008	+0.050	+0.013	-0.003
ELECTRA-large	0	10.20%	-0.005	+0.024	+0.007	-0.000
RoBERTa-base	0	11.77%	-0.007	+0.030	+0.008	-0.003
RoBERTa-large	0.3	5.18%	-0.001	+0.014	-0.001	-0.001
DeBERTa-v2-320M	0.3	6.53%	+0.001	+0.015	-0.002	-0.003
ELECTRA-large	0.3	3.68%	-0.000	+0.009	-0.000	-0.002

Two decision rules over one set of probabilities, on the pdf_group protocol. M0 takes each field’s argmax independently; M1 projects onto the 17 legal states top-down; M4 scores all 17 and takes the best. **M1 and M4 both emit an invalid tuple on 0% of rows in every arm** – their output space is the legal set – so the invalid column reports M0 only, where it ranges from 3.68% to 19.75%. Means over 3 seeds; paired PDF-cluster bootstrap, 10,000 resamples, seed 20260814, one resample shared within each contrast. Enforcing legality raises whole-row accuracy in 7 of 7 arms, 7 of those intervals excluding zero. On the official metric it is negative in 4 of the 4 arms trained without the structural objective and detectable in 3 of them, against 0 of 3 among the structurally trained arms. Joint decoding lowers whole-row accuracy relative to projection in 7 of 7 arms, detectable in 2; on the official metric it is detectable in 3 of the 4 untrained arms and 0 of the 3 trained ones. Each of those 3 detectable cells is positive. **Exploratory.** These contrasts were named after the primary analysis and form no Holm family; bold marks an interval excluding zero, not a corrected verdict. The claim is the sign pattern across arms, not any single cell. See docs/governance/inference_families.md.

table4_contrasts

Metric	Contrast	Δ	95% CI	p_{Holm}
Weighted macro-F1 (official)	M1-M0	-0.001	[-0.006, 0.003]	1.000
	M3-M2	+0.001	[-0.004, 0.006]	1.000
	M4-M1	+0.000	[-0.005, 0.005]	1.000
	M6-M3	-0.008	[-0.018, 0.002]	0.416
	M6-M5	-0.008	[-0.016, -0.001]	0.171
Tuple accuracy	M1-M0	+0.035	[0.028, 0.043]	0.001
	M3-M2	+0.004	[-0.001, 0.008]	0.370
	M4-M1	-0.006	[-0.010, -0.002]	0.028
	M6-M3	-0.005	[-0.015, 0.004]	0.505
	M6-M5	-0.004	[-0.011, 0.004]	0.505

The five pre-specified contrasts of the analysis plan, under the two metrics the plan named before any result existed: the competition’s own weighted macro-F1 and tuple exact-match. Paired PDF-cluster bootstrap over 10,000 resamples on the document-disjoint protocol; the resampling unit is the source report, not the paragraph. Each metric is Holm-corrected as its own family of five rather than pooled: the contrasts were specified once, not once per metric. **Bold p_{Holm} marks a contrast that survives the correction; the bracketed intervals are uncorrected 95% percentile intervals and decide nothing on their own.** Two further structure-aware metrics were examined after the primary analysis and are reported as exploratory in the text, not as Holm families here. The largest effect on the official metric, M6-M5 at $|\Delta| = 0.008$, has an interval excluding zero and still does not survive the correction ($p_{\text{Holm}} = 0.171$) – and those two rules are the highest- and lowest-scoring of the seven, so that is not one contrast among many but the entire spread the benchmark has to resolve. Of the five contrasts, none on Weighted macro-F1 (official) and two on Tuple accuracy survive the correction at $\alpha=0.05$. The official metric and tuple accuracy were named in the analysis plan in advance; the path-constrained variant and the hierarchical F1 were adopted after the primary analysis and are not pre-specified.

table5_headroom

Field	Weight	macro-F1	Shortfall	Share	Value of +0.1 per class
evidence_quality	0.35	0.418	0.204	47.5%	0.0087
evidence_status	0.30	0.651	0.105	24.4%	0.0100
verification_timeline	0.15	0.467	0.080	18.6%	0.0030
promise_status	0.20	0.796	0.041	9.5%	0.0100

Where the official score falls short under M1, and what closing each part is worth. Shortfall is the field’s weight times its distance from a perfect macro-F1; share is its portion of the total. The last column is what a 0.1 gain on a single class’s F1 adds to the weighted total – the field’s weight divided by its class count, times 0.1 – which orders the fields differently from the shortfall: macro-F1 averages over classes, so a field carrying more of them returns less per class improved. **Not all of the shortfall is reachable.** Misleading occurs twice in the development set and is never predicted by any arm, so a quarter of evidence_quality’s macro-F1 – 0.0875 of the weighted total – is out of reach by construction.

table6_regimes

Method	Same-document	Document-disjoint	Δ	95% CI
M0	0.585	0.572	0.012	[0.004, 0.022]
M1	0.581	0.571	0.010	[0.001, 0.019]
M2	0.587	0.574	0.012	[0.001, 0.024]
M3	0.588	0.575	0.012	[0.002, 0.023]
M4	0.586	0.571	0.014	[0.005, 0.024]
M5	0.590	0.576	0.015	[0.004, 0.027]
M6	0.587	0.567	0.020	[0.011, 0.030]

Same-document versus document-disjoint evaluation. The left column measures seen-report, unseen-paragraph generalisation and matches the competition distribution, since test and development draw on the same 49 reports; the right column measures generalisation to entirely unseen reports. Δ is the gap between two estimation targets and is not a bias estimate. All seven decision rules are reported rather than a best-scoring subset: every Δ is positive and every interval excludes zero, so the gap is a property of the split and not of any one rule.

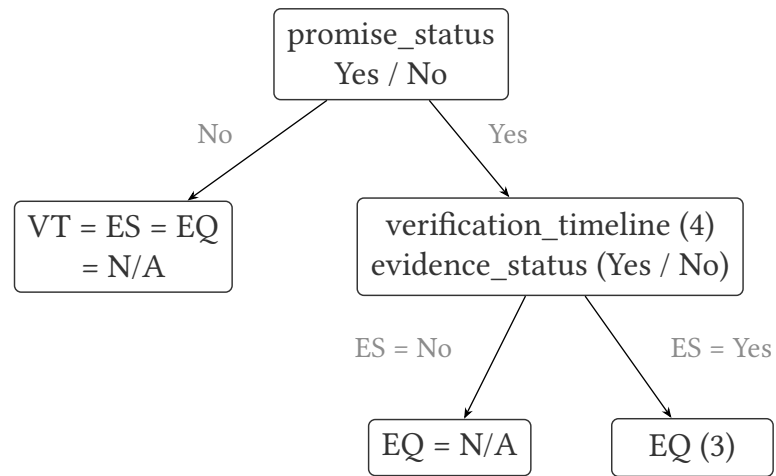
table7_multilingual_mechanism

Corpus	Rows	M0 illegal %	<i>N/A</i> net	Subst. net	Δ official	Δ whole-row
Chinese (AI CUP)	2,000	12.55	+395	-264	-0.0011	+0.0350
English	400	23.17	+148	-86	+0.0032	+0.0725
French	400	21.08	+150	-107	+0.0036	+0.0625
Japanese	400	31.25	+183	-245	-0.0131	+0.0708
Korean	500	22.73	+103	-148	-0.0192	+0.0293

Projection (M1) against independent argmax (M0) on five corpora, each on its own primary backbone at $\lambda = 0$ and its document-disjoint protocol, means over three seeds. M0's illegal rate spans 12.55% to 31.25%; M1's is 0% everywhere by construction. The two net columns are the mechanism. Projection's only lever on a child field is writing *N/A*, so its repairs land on that class and its damage on the substantive ones: the sign pattern is *N/A* positive and substantive negative in five of five corpora. Macro-F1 weights those classes equally, which is where the movement cancels. Whole-row accuracy rises in five of five corpora; the official metric rises in two. The same predictions, scored two ways, do not agree on whether anything improved. **The metric's exchange rate is not the same in every corpus.** macro-F1 averages over the classes present in gold, and Korean has no Misleading row, so its evidence_quality divides by one class fewer: a point of one class's F1 is worth 0.1167 there against 0.0875 elsewhere. Cross-corpus scores are therefore not on one scale and are not ranked here.

figure1_hierarchy

(a) Task hierarchy



$1 + 4 \times (1 + 3) = 17$ legal tuples

of $2 \times 5 \times 3 \times 4 = 120$ combinations

the other 103 are hierarchy-invalid

(b) Decision routes (alternatives, not stages)

